Deep Learning for Renewable Energy: A Comprehensive Review

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ABSTRACT

Deep learning has become an important technology in renewable energy forecasting. This review surveys recent advances in neural network architectures including convolutional and trans- former models. We also discuss data quality challenges and the use of attention mechanisms for multivariate time series.

Keywords: deep learning; renewable energy; time series; attention

1 Introduction

Deep learning has become an important technology in renewable energy. In this paper we review the litera- ture on neural network based forecasting. The field has grown rapidly since 2015. Several surveys ex- ist, but none focuses on infor- mation flow between modules.

2 Methods

We group methods into three families. The first family uses convolutional architectures. The second family relies on recurrent units such as LSTM. The third family is based on the trans- former architecture with self-attention.

2.1 Data Preparation

Input series are normalized to zero mean. Missing values are imputed by linear interpolation before model training.

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Figure 1. Overview of the deep learning pipeline for energy forecasting. The encoder maps raw series into latent features.

Table 1. Comparison of major architectures

Method	Accuracy	Year
CNN	0.91	2019
LSTM	0.89	2018
Transformer	0.94	2021

3 Results and Discussion

Transformers achieve the best accuracy on most benchmarks. However they require more data. Hybrid mod- els combine convolutional feature extractors with attention layers and show robust performance.

4 Challenges

Data scarcity remains a major barrier. Model interpretability is another open problem. Standardized benchmarks are needed.

5 Conclusion

We reviewed deep learning methods for renewable energy forecasting and outlined future directions.

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